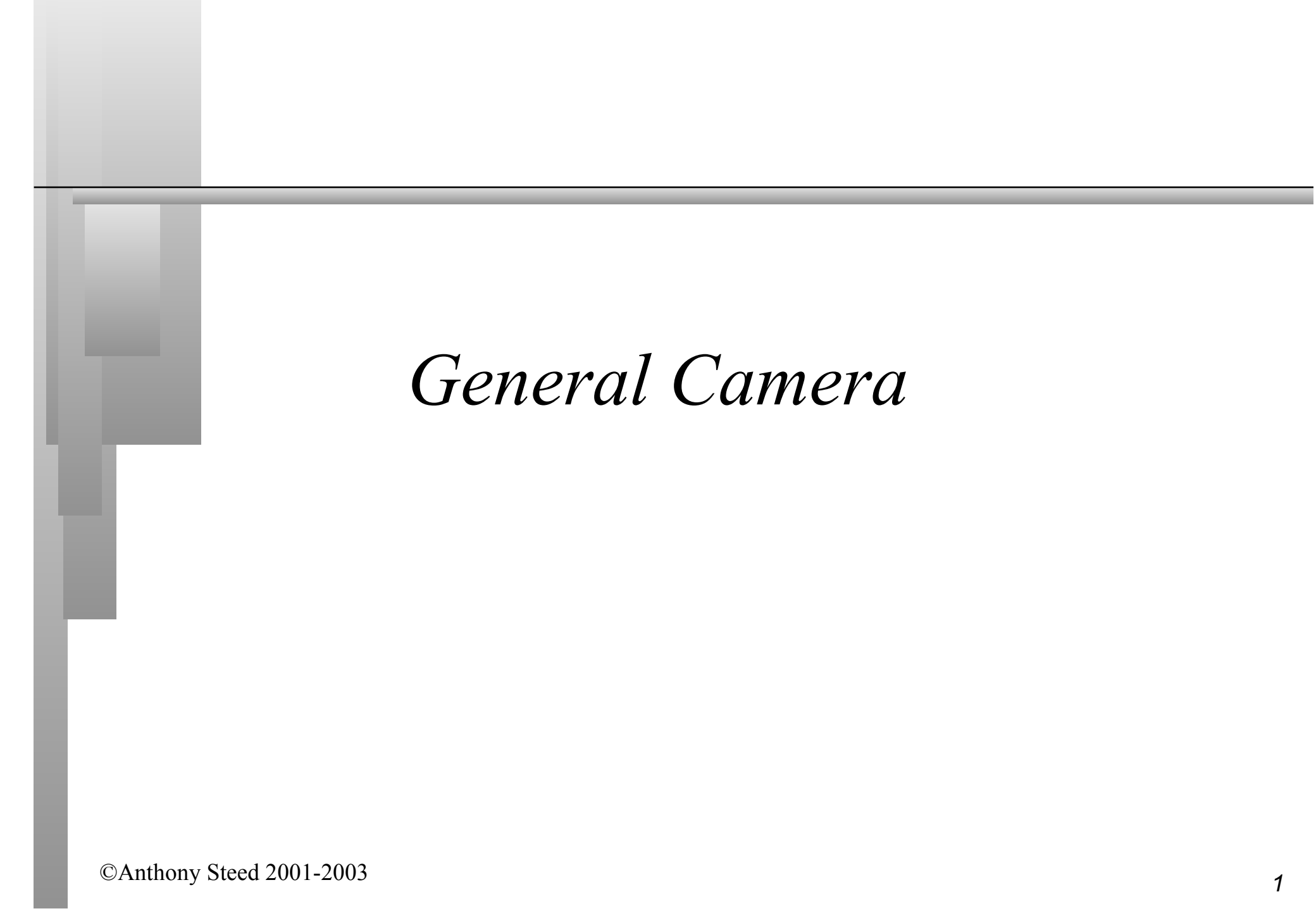


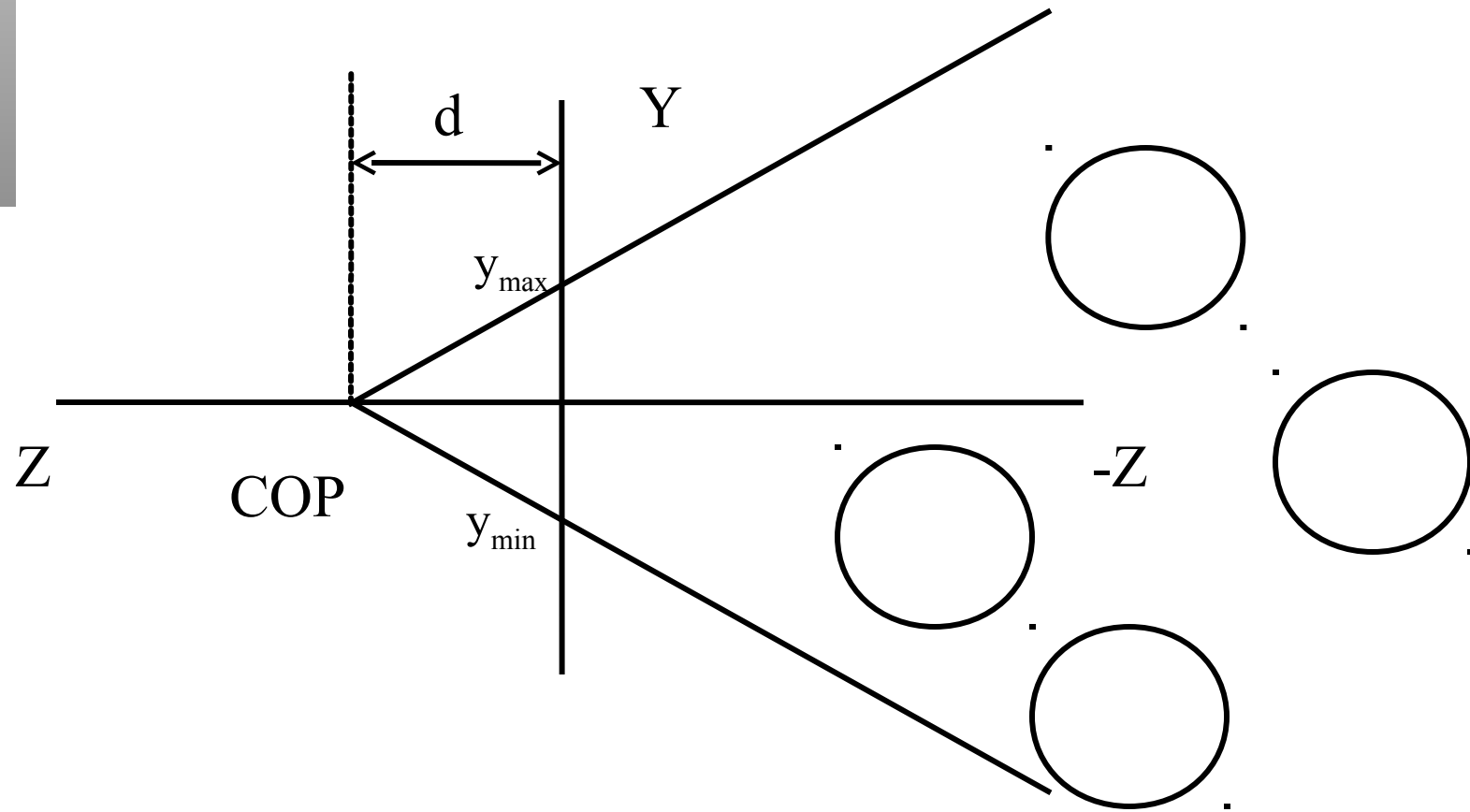
A decorative graphic consisting of a thick gray vertical bar on the left side of the slide and a thin gray horizontal line that spans the width of the slide, intersecting the vertical bar.

General Camera

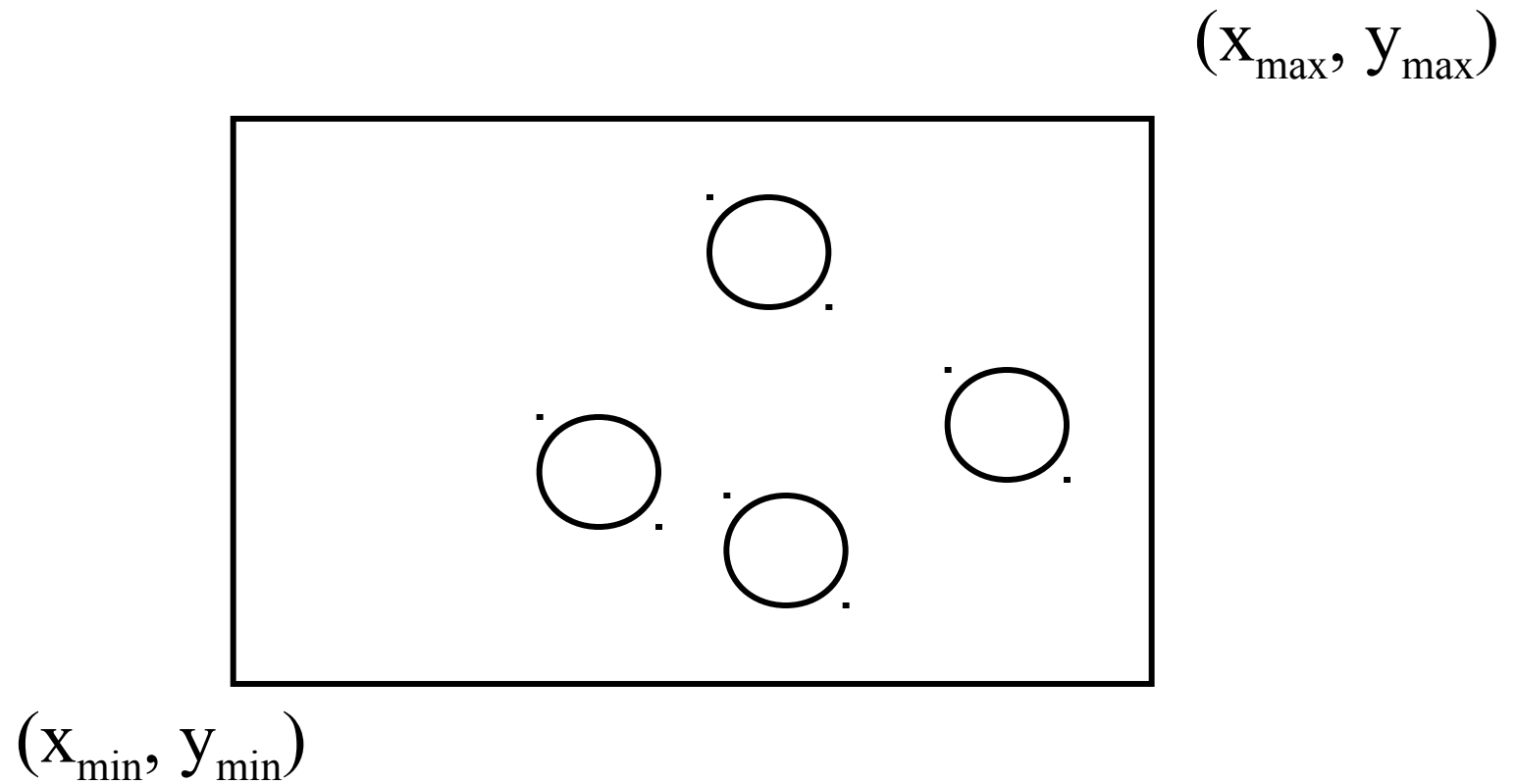
Overview

- *Simple camera is limiting and it is necessary to model a camera that can be moved*
- *We will define parameters for a camera in terms of where it “is”, the direction it points and the direction it considers to be “up” on the image*

Simple Camera (Cross Section)



View From the Camera



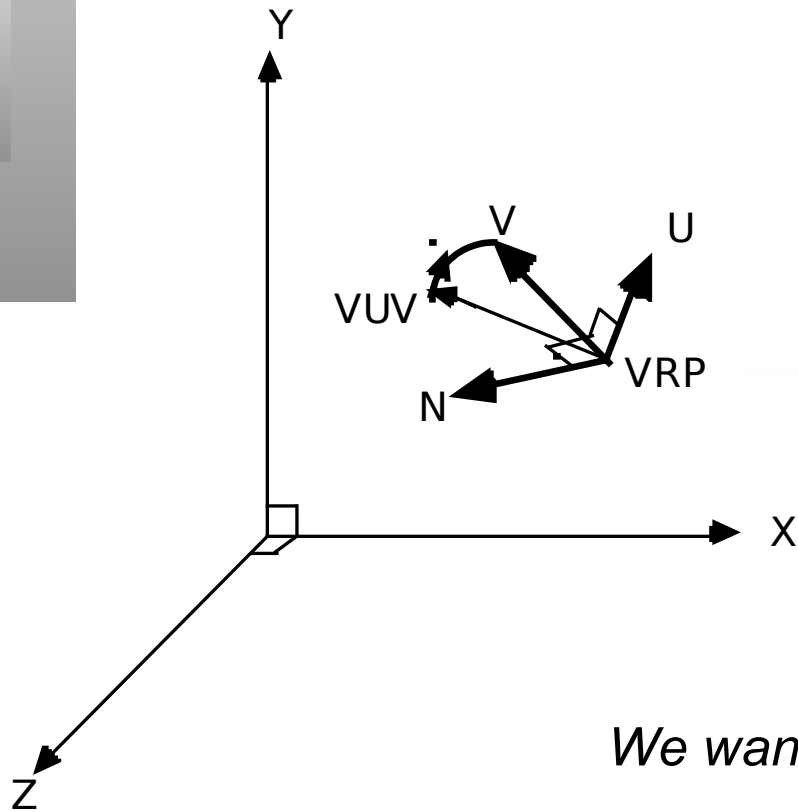
General Camera

- *View Reference Point (VRP)*
 - *where the camera is*
- *View Plane Normal (VPN)*
 - *where the camera points*
- *View Up Vector (VUV)*
 - *which way is up to the camera*
- *X (or U-axis) forms LH system*

UVN Coordinates

- *View Reference Point (VRP)*
 - *origin of VC system (VC=View Coordinates)*
- *View Plane Normal (VPN)*
 - *Z (or N-axis) of VC system*
- *View Up Vector (VUV)*
 - *determines Y (or V-axis) of VCS*
- *X (or U-axis) forms Left Hand system*

World Coords and Viewing Coords

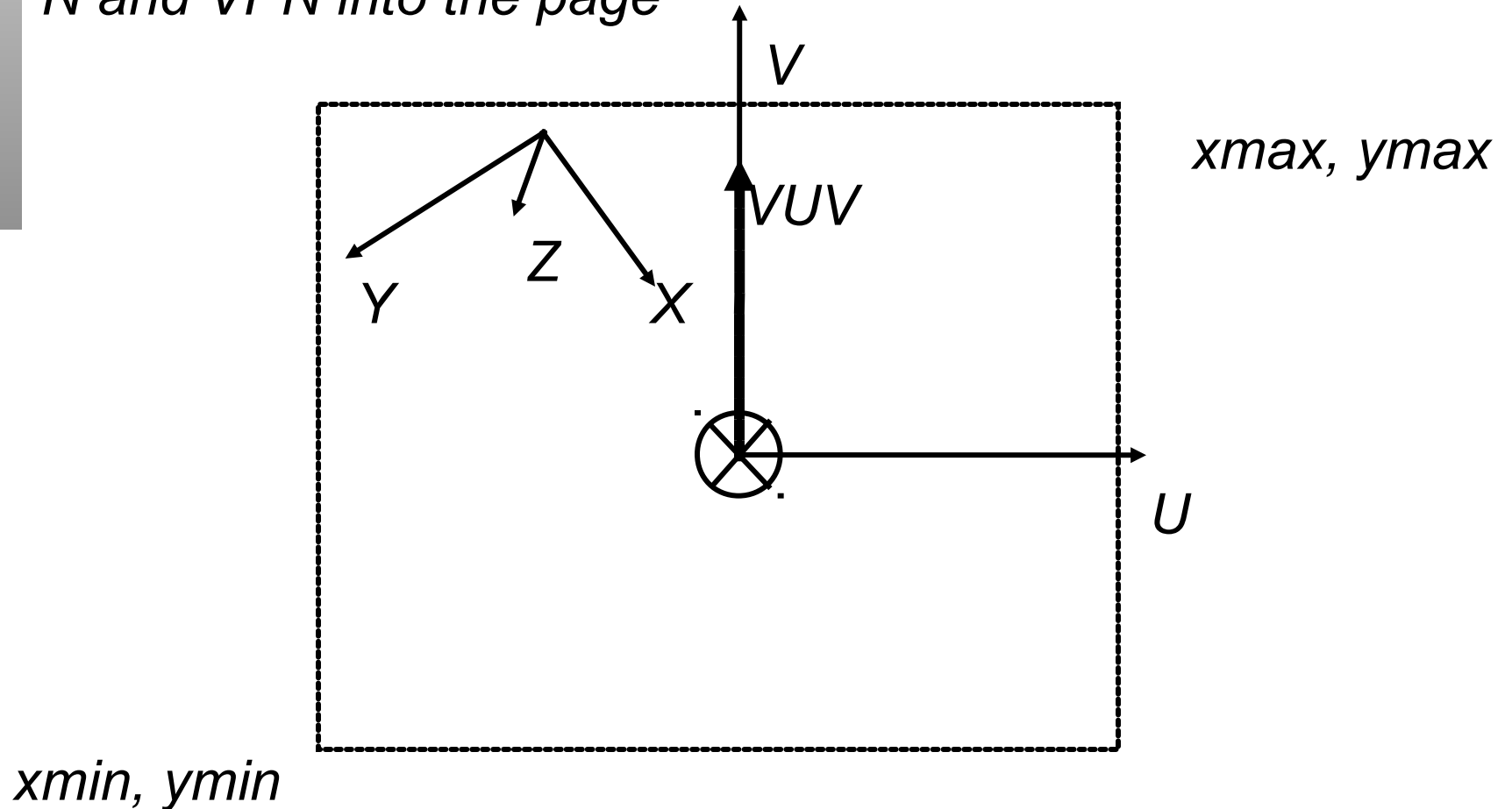


$$\begin{pmatrix} R_1 & R_2 & R_3 & 0 \\ R_4 & R_5 & R_6 & 0 \\ R_7 & R_8 & R_9 & 0 \\ T_1 & T_2 & T_3 & 1 \end{pmatrix}$$

We want to find a general transform (EQ1) of the above form that will map WC to VC

View from the Camera

N and VPN into the page



Finding the basis vectors

- *Step 1 - find n* $n = \frac{V \times P \times N}{|V \times P \times N|}$
- *Step 2 - find u* $u = \frac{n \times V \times U}{|n \times V \times U|}$
- *Step 3 - find v* $v = u \times n$

Finding the Mapping (1)

- *u, v, n must rotate under R to i, j, k of viewing space*

$$\begin{pmatrix} u \\ v \\ n \end{pmatrix} \begin{pmatrix} R \end{pmatrix} = \begin{pmatrix} I \end{pmatrix} \quad R = \begin{pmatrix} u_1 & v_1 & n_1 \\ u_2 & v_2 & n_2 \\ u_3 & v_3 & n_3 \end{pmatrix}$$

- *Both basis are normalised so this is a pure rotation matrix*
 - *recall in this case $R^T = R^{-1}$*

Finding the Mapping (2)

- *In uvn system VRP (q) is (0 0 0 1)*
- *And we know from EQ1 $qR + t = 0$ so*

$$\begin{aligned} t &= -qR \\ &= -\left(\sum_{i=1}^3 q_i u_i \quad \sum_{i=1}^3 q_i v_i \quad \sum_{i=1}^3 q_i n_i \right) \end{aligned}$$

Complete Mapping

■ Complete matrix

$$M = \begin{pmatrix} u_1 & v_1 & n_1 & 0 \\ u_2 & v_2 & n_2 & 0 \\ u_3 & v_3 & n_3 & 0 \\ -\sum_{i=1}^3 q_i u_i & -\sum_{i=1}^3 q_i v_i & -\sum_{i=1}^3 q_i n_i & 1 \end{pmatrix}$$

For you to check

■ *If*

$$M = \begin{pmatrix} R & 0 \\ -qR & 1 \end{pmatrix}$$

■ *Then*

$$M^{-1} = \begin{pmatrix} R^T & 0 \\ q & 1 \end{pmatrix}$$

Using this for Ray-Casting

- *Use a similar camera configuration (COP is usually, but not always on -n)*
- *To trace object must either*
 - *transform spheres into VC*
 - *transform rays into WC*

Ray-casting

- *Transforming rays into WC*
 - *Transform end-point once*
 - *Find direction vectors through COP as before*
 - *Transform vector by $\begin{pmatrix} R^T & 0 \\ q & 1 \end{pmatrix}$*
 - *Intersect spheres in WC*

Ray-casting

- *Transforming spheres into VC*
 - *Centre of sphere is a point so can be transformed as usual (WC to VC)*
 - *Radius of sphere is unchanged by rotation and translation (and spheres are spheroids if there is a non-symmetric scale)*

Tradeoff

- *If more rays than spheres do the former*
 - *transform spheres into VC*
- *For more complex scenes e.g. with polygons*
 - *transform rays into WC*

Alternative Forms of the Camera

■ *Simple “Look At”*

- *Give a VRP and a target (TP)*
- *$VPN = TP - VRP$*
- *$VUV = (0 \ 1 \ 0)$ (i.e. “up” in WC)*

■ *Field of View*

- *Give horizontal and vertical FOV or one or the other and an aspect ratio*
- *Calculate viewport and proceed as before*

Animated Cameras

- *Animate VRP (observer-cam)*
- *Animate VPN (look around)*
- *Animate TP (track-cam)*
- *Animate COP*
 - *along VPN - zoom*
 - *orthogonal to VPN - distort*

Recap

- *We created a more general camera which we can use to create views of our scenes from arbitrary positions*
- *Formulation of mapping from WC to VC (and back)*